

## Number, units and estimation - Set 2

TMUA practice questions by topic. Specification M1.1, M1.2, M2.1-M2.14. 9 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 Water flows through a pipe at 1.2 litres per second. How long does it take to fill a tank of volume  $3.6 \text{ m}^3$ ? ( $1 \text{ m}^3 = 1000 \text{ litres}$ .)
- A 30 minutes  
B 50 minutes  
C 5 minutes  
D 1 hour  
E 3 hours  
F 500 seconds
- 2  $a = 5.0$  and  $b = 2.0$ , each correct to 2 significant figures. Find the lower bound of  $\frac{a}{b}$ .
- A  $\frac{4.5}{2.5}$   
B  $\frac{4.5}{2.05}$   
C  $\frac{4.05}{2.05}$   
D  $\frac{4.05}{2.0}$   
E  $\frac{2.4}{5.05}$   
F  $\frac{2.05}{2.05}$
- 3 What is the smallest positive integer that is divisible by every integer from 1 to 10?
- A 3628800  
B 1260  
C 840  
D 720  
E 5040  
F 2520

- 4 Which of the following fractions has a terminating decimal expansion?
- A  $\frac{7}{40}$
  - B  $\frac{9}{22}$
  - C  $\frac{11}{14}$
  - D  $\frac{14}{9}$
  - E  $\frac{5}{12}$
- 5 Find the exact value of  $(\sqrt{5} - 1)^2 + (\sqrt{5} + 1)^2$ .
- A 12
  - B 8
  - C 10
  - D 6
  - E  $12 + 4\sqrt{5}$
  - F  $4\sqrt{5}$
- 6 The lowest common multiple of 12, 18 and  $n$  is 180. Which of the following could be  $n$ ?
- A 20
  - B 24
  - C 8
  - D 40
  - E 50
  - F 27
- 7 How many positive divisors does 360 have?
- A 24
  - B 20
  - C 18
  - D 36
  - E 12
  - F 16

- 8**  $n$  is a positive integer such that  $n$ ,  $n + 2$  and  $n + 4$  are all prime. Which of the following statements is/are true?

- I** There is exactly one such value of  $n$ .
- II**  $n$  must be odd.
- III** One of  $n$ ,  $n + 2$ ,  $n + 4$  must be divisible by 3.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

- 9**  $n$  is a positive integer greater than 1. Which of the following statements is/are true?

- I** If  $n$  is not prime, then  $n$  has a prime factor at most  $\sqrt{n}$ .
- II** If  $n$  has no prime factor at most  $\sqrt{n}$ , then  $n$  is prime.
- III** If  $n$  is prime, then  $n$  has no factor between 1 and  $n$  exclusive.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III